

CS 486/686

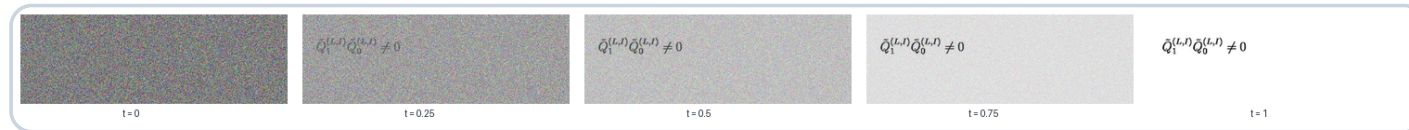
Dissecting Diffusion and World Models

Yuntian Deng

Lecture 24

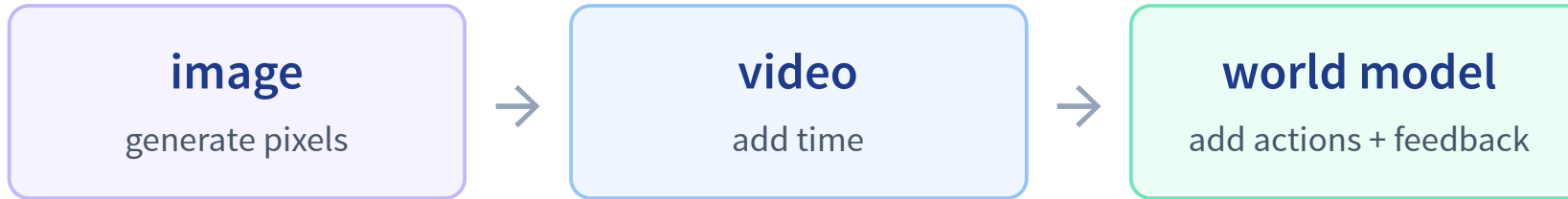
From noise to video to an interactive world

From noise to an image



[Deng et al., “Latent Alignment and Variational Attention,” 2017.](#)

Image → video → interactive computer



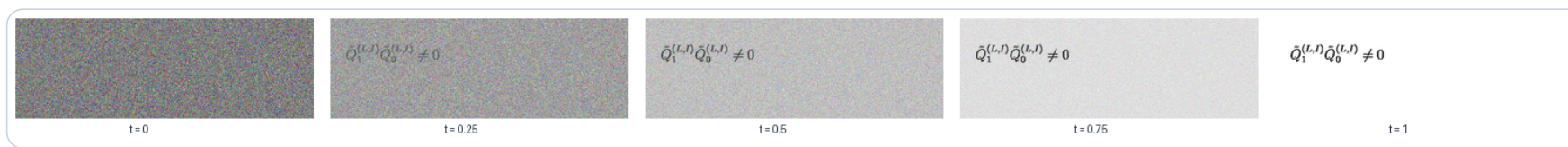
Turn each unfamiliar problem into one we already know how to solve.

Start with noise and an image



Generation learns how to travel from the left endpoint to the right.

Connect them with a path INTERACTIVE



$$z_t = (1 - t)z_{\text{noise}} + tz_{\text{data}}$$

Velocity is the slope of the path

$$z_t = (1 - t)z_{\text{noise}} + tz_{\text{data}}$$

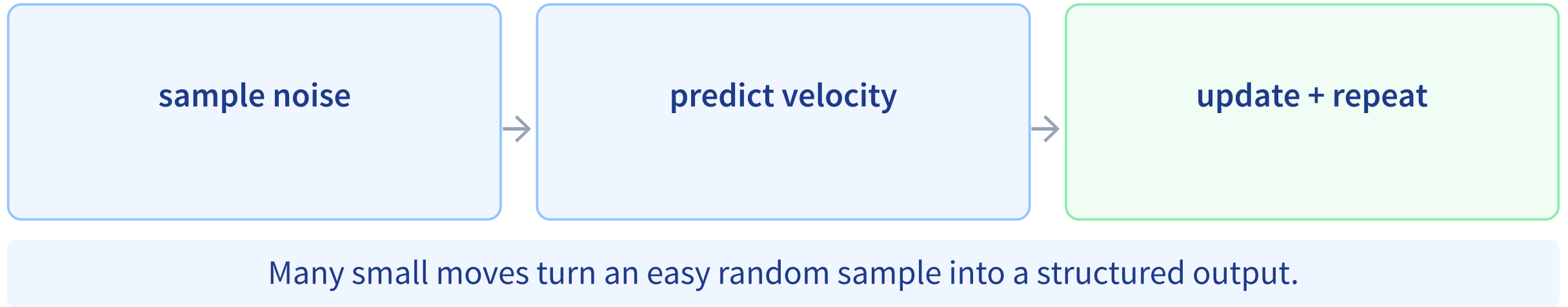
↓ differentiate with respect to t

$$v^* = \frac{dz_t}{dt} = z_{\text{data}} - z_{\text{noise}}$$

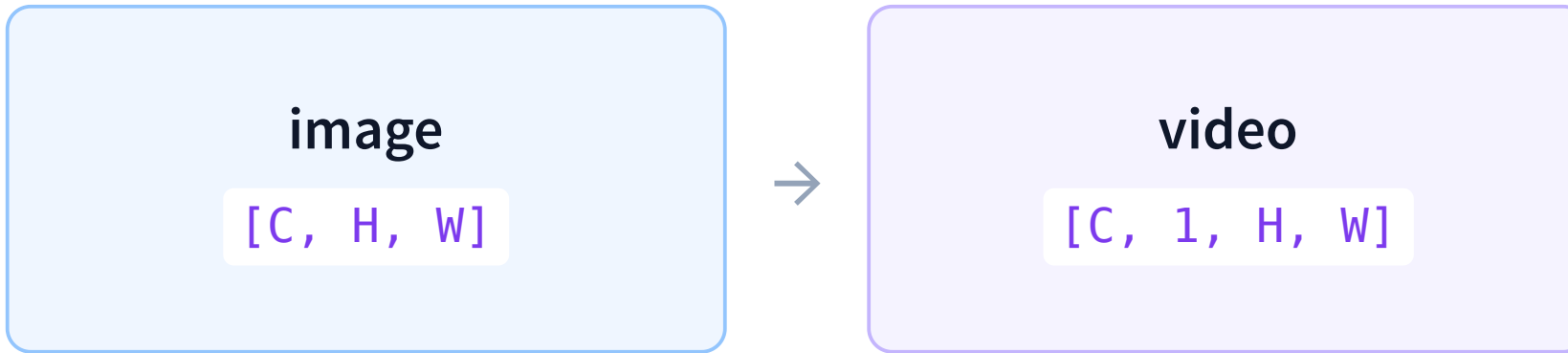
distance left: 

slope: 

Generation follows those arrows repeatedly

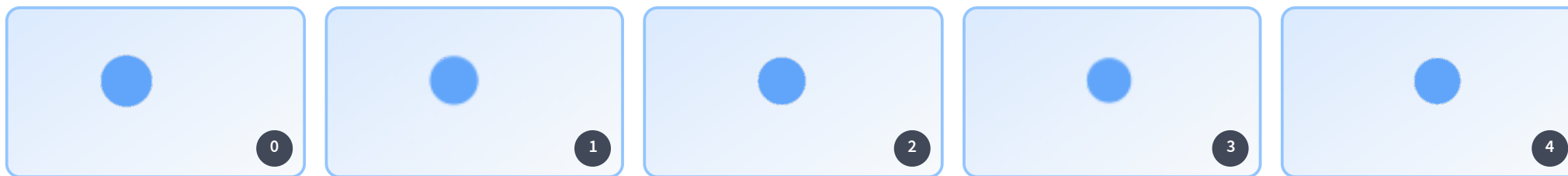


An image is a one-frame video



Add one axis: time.

Video adds time



same object

smooth motion

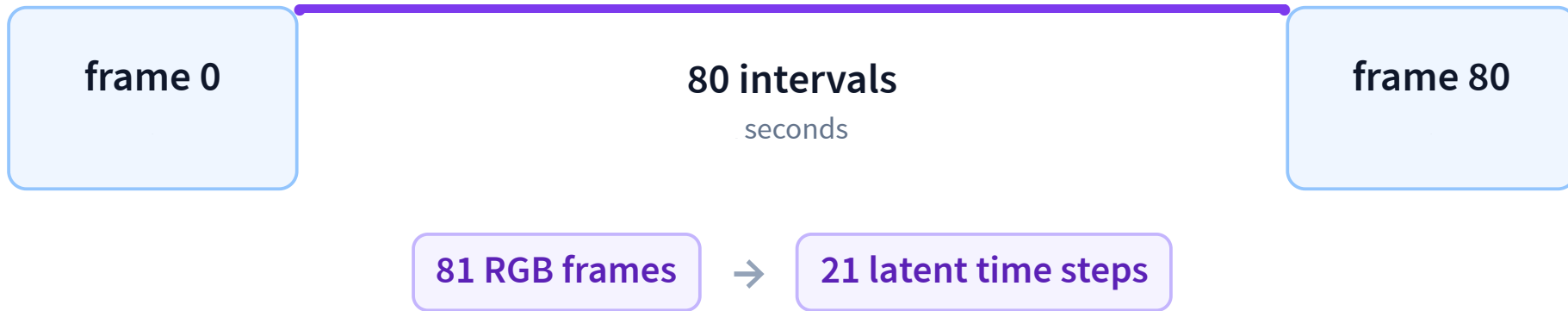
consistent scene

Five seconds is about 80 frames

$$5 \text{ seconds} \times 16 \text{ frames/second} = 80 \text{ frame intervals}$$

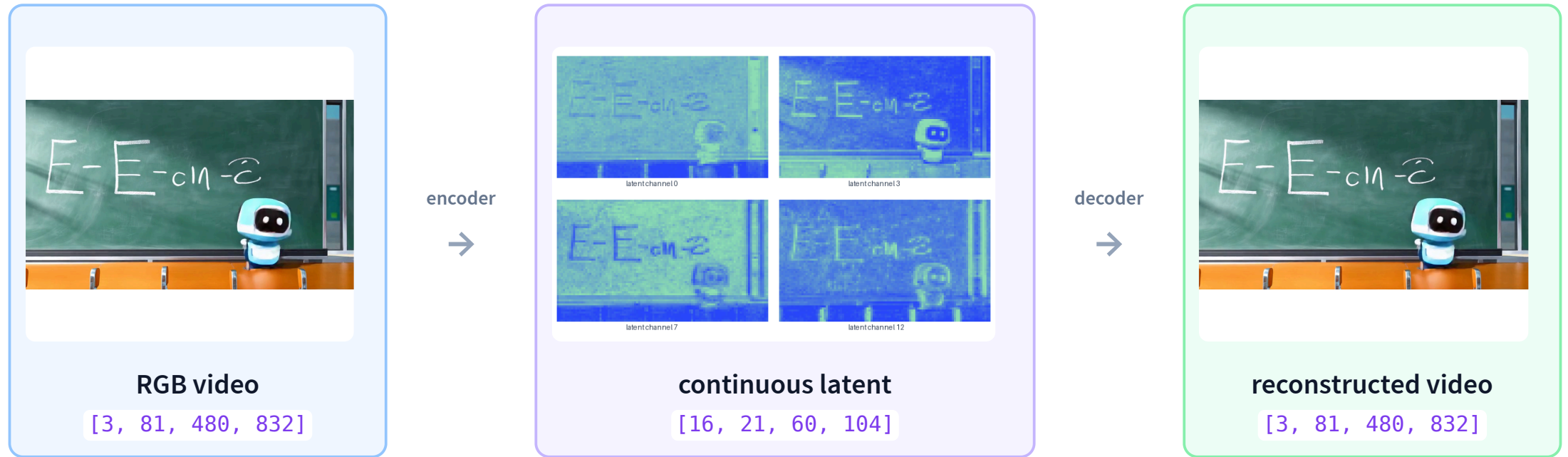
This is the right scale for intuition. Wan's exact default has one extra endpoint frame.

Wan generates 81 frames by default



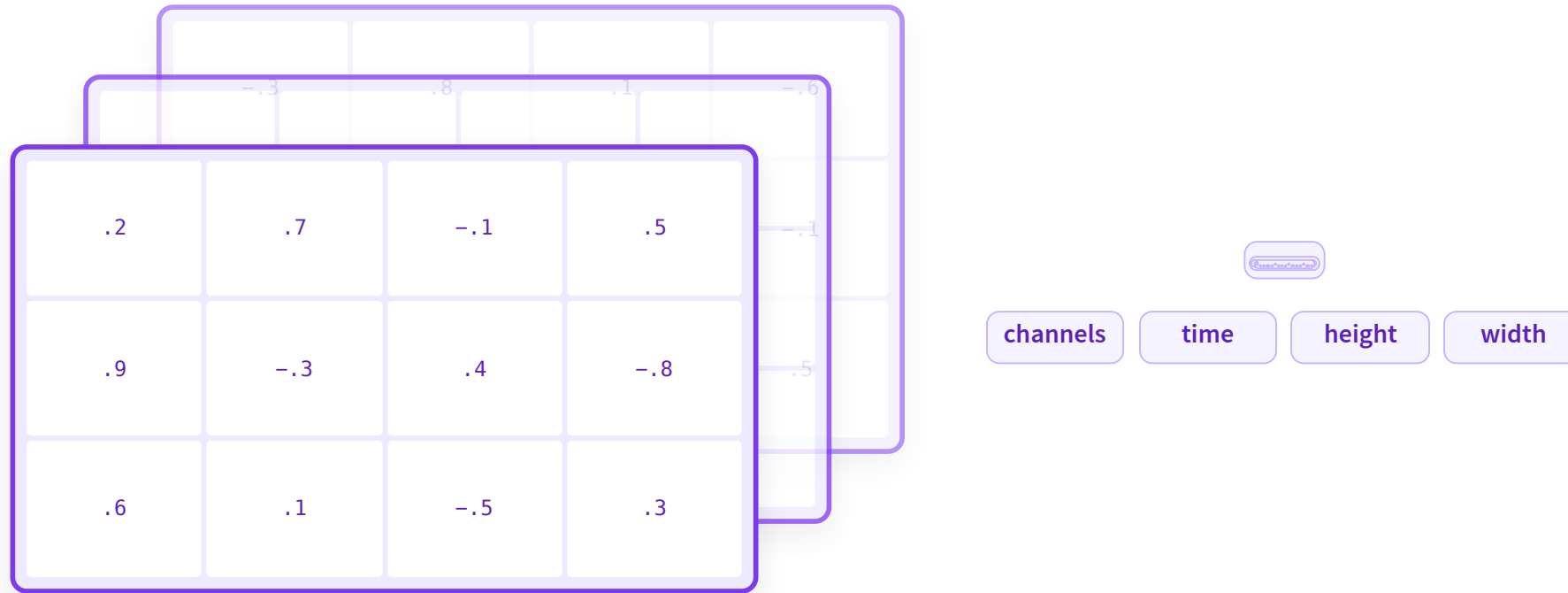
Other valid lengths follow $4k + 1$, such as 33, 49, or 81 frames.

The Wan-VAE compresses the video



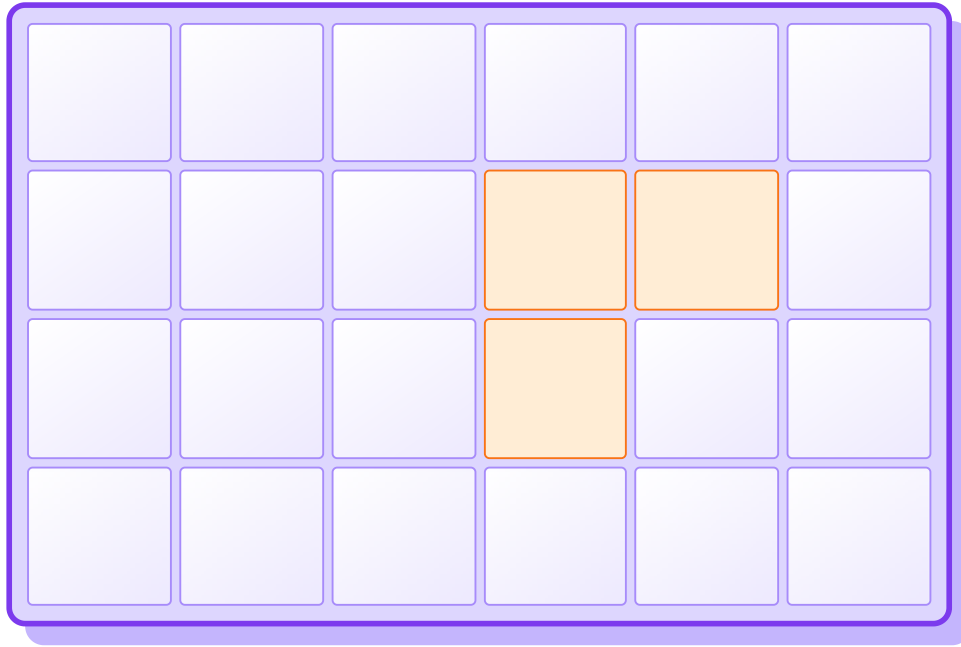
Real Wan-VAE encode/decode example; [provenance and hashes](#).

The latent is a 4D array



No vocabulary and no token IDs—just a compact multi-dimensional array of real numbers.

Split the latent grid into patches



patch size

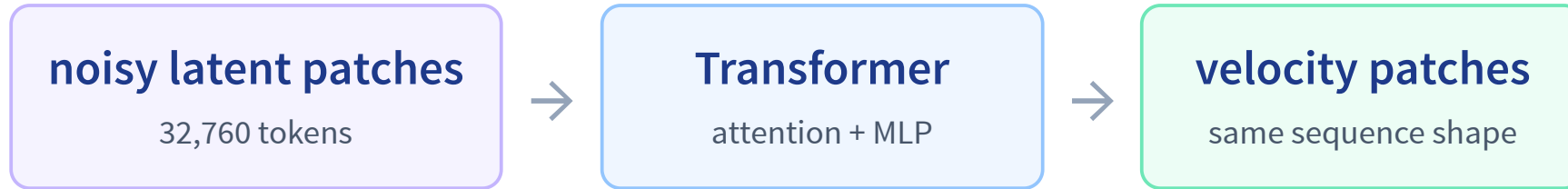
in latent time, height, width

sequence length

token width

each patch projects to 1536 numbers

A DiT is a transformer for noisy latent patches



Peebles & Xie, [“Scalable Diffusion Models with Transformers”](#), ICCV 2023.

Wan2.1-T2V-1.3B at a glance

Wan2.1

T2V-1.3B

Wan-AI/Wan2.1-T2V-1.3B

1.3B

DiT parameters

30

DiT blocks

1536

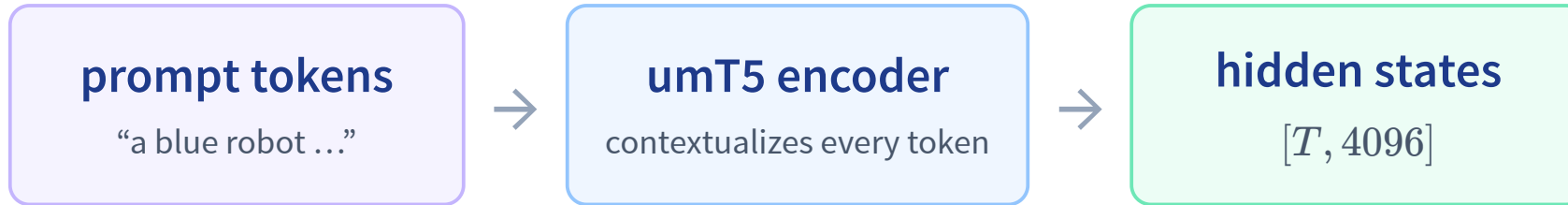
hidden width

480p

target resolution

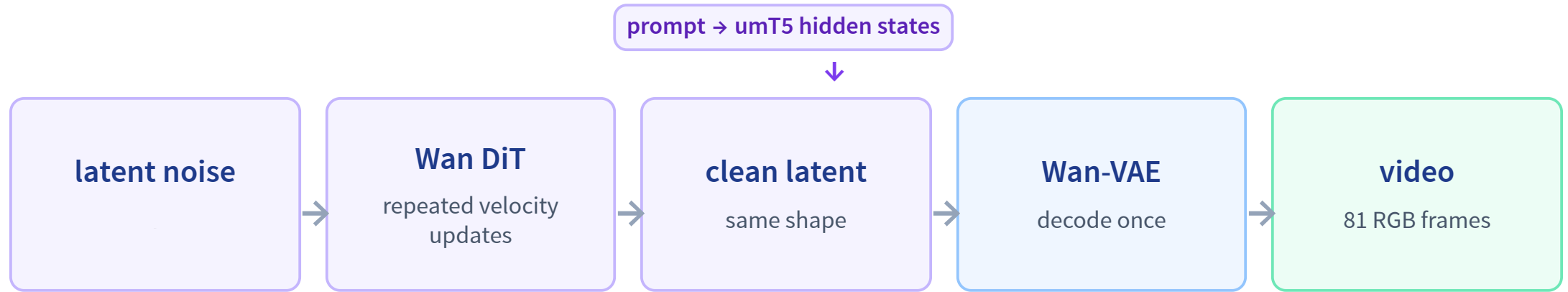
[Wan technical report](#) and official checkpoint configuration.

umT5 turns the prompt into vectors

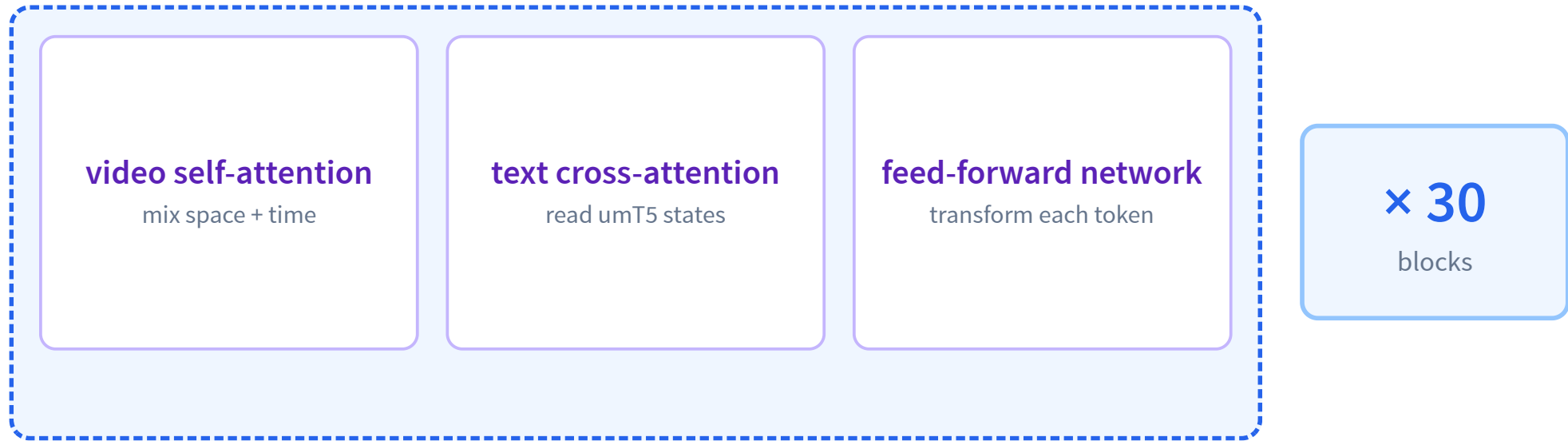


The DiT reads contextual vectors—not raw words.

Prompt in; video out

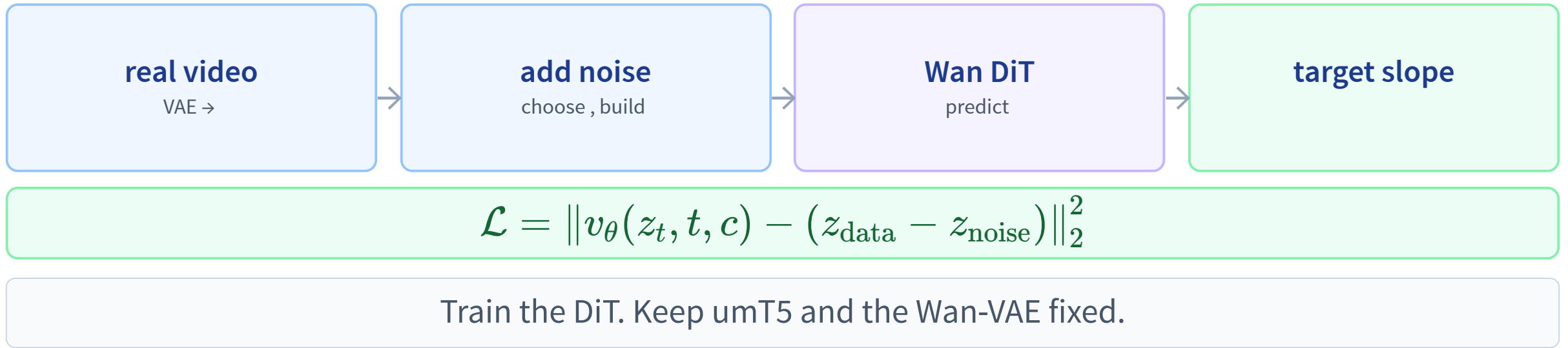


A Wan block mixes video, text, and time



The timestep modulates each block, telling it how noisy the current latent is.

Train the DiT to predict velocity



Generate by following the predicted velocity

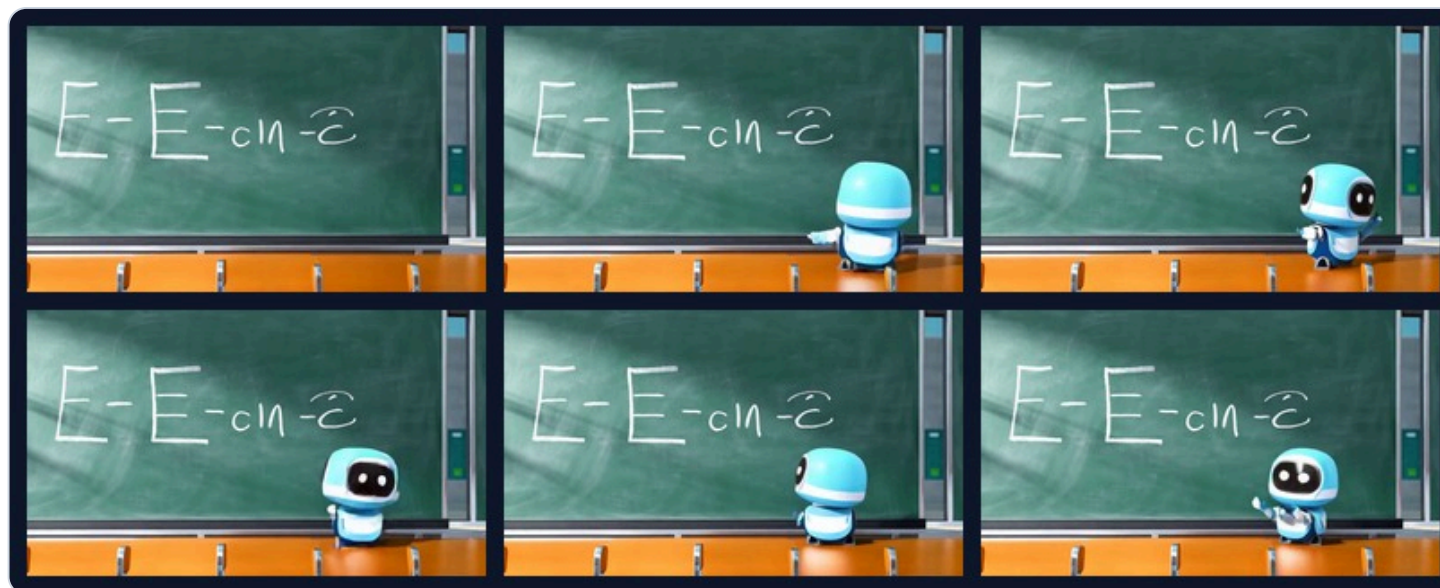


The generated result

RECORDED MODEL OUTPUT

PROMPT

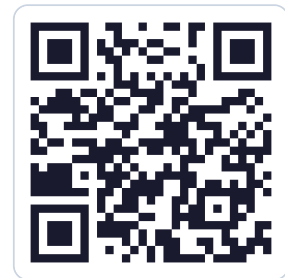
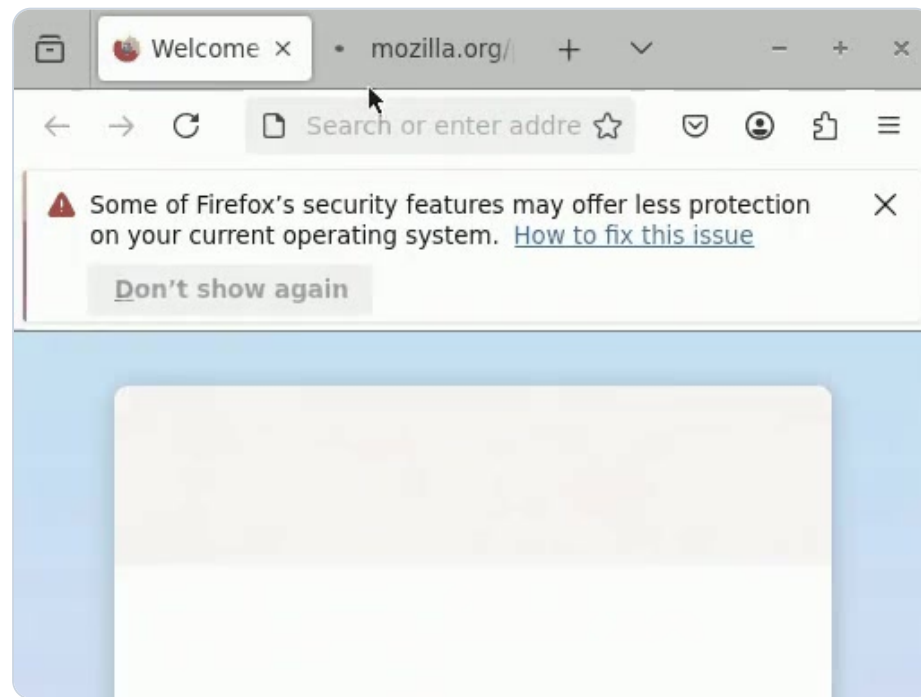
A small blue robot writes $E = mc^2$ on a blackboard in a sunlit university classroom, then turns toward the camera and waves, fixed camera, coherent motion, clean cinematic lighting.



Generated with [Wan2.1-T2V-1.3B](#); [reproducibility record](#).

What if the whole computer were a neural network?

Mouse and keyboard go in. The next screen is generated.



neural-os.com

[Deng et al., "NeuralOS," 2025.](#)

A video generation model is not a world model

video generator

prompt once

fixed clip

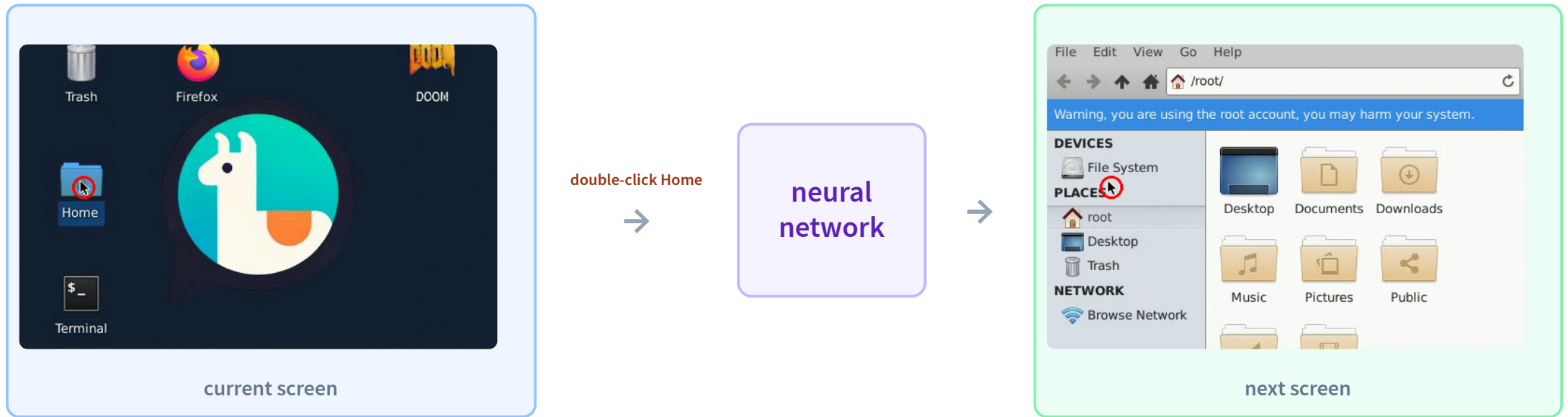
world model

observe each action

predict → feed back ↺

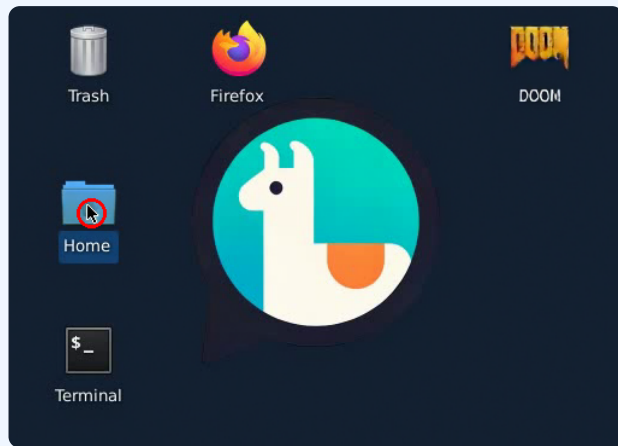
A generated video is static. You cannot interact with it like a video game.

NeuralOS predicts the next screen

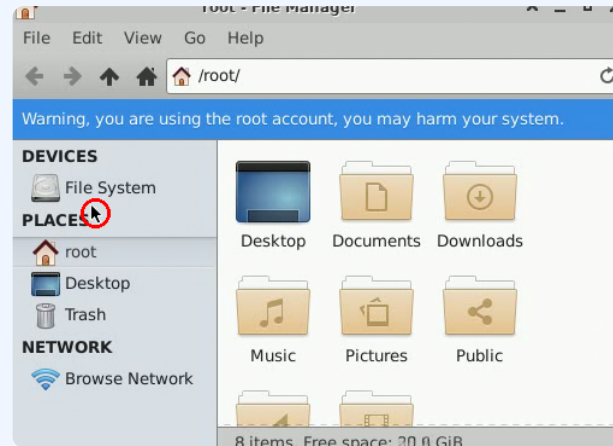


(screen history, action) → next screen

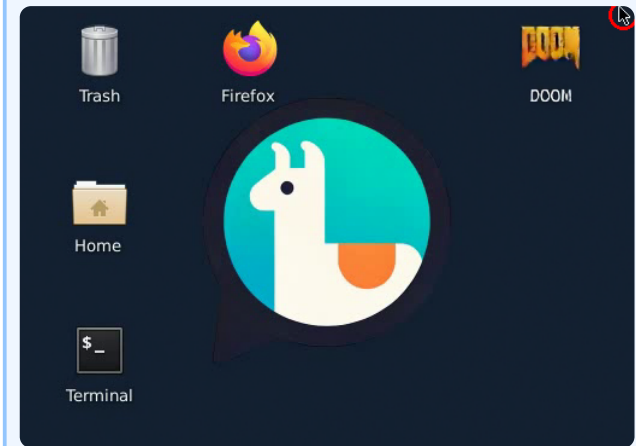
Feed the prediction back and repeat STEP THROUGH



double-click Home



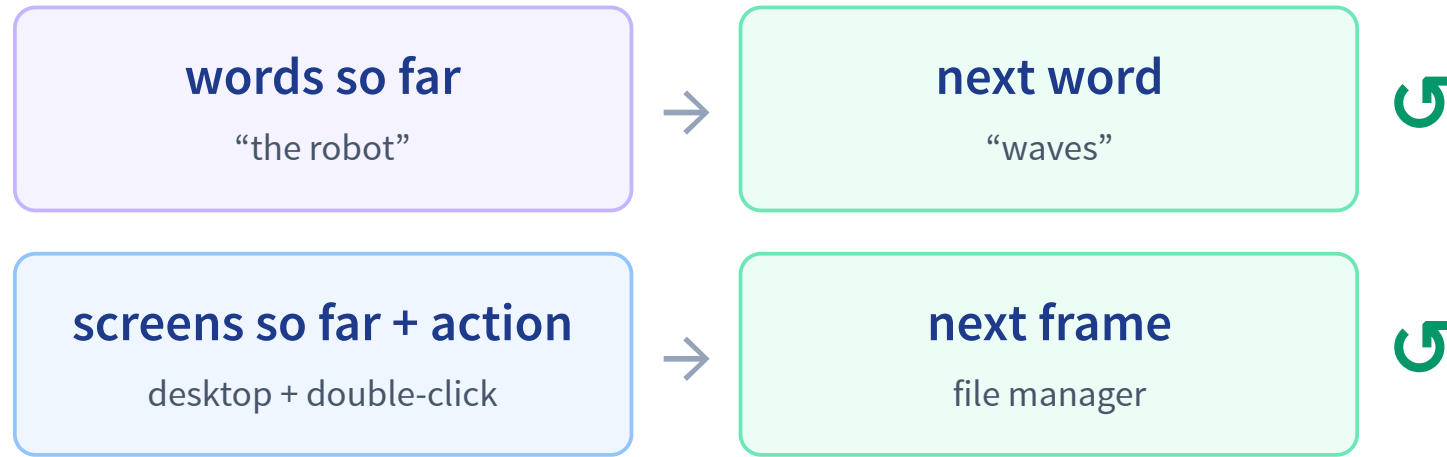
generated



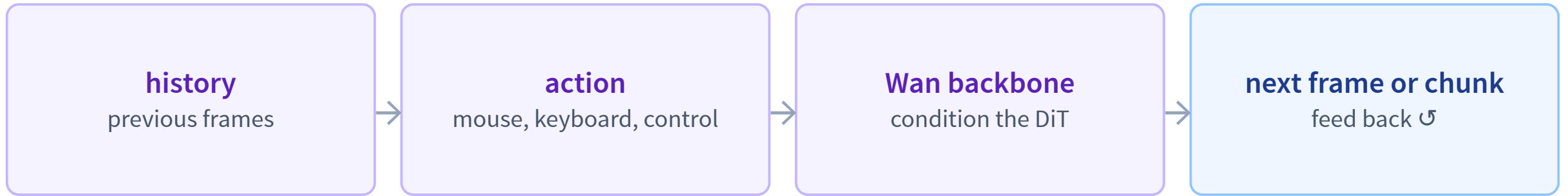
click x → generated

The user can choose a new action between predictions.

Next word prediction → next frame prediction



Wan can be adapted to predict what happens next



NeuralOS uses a different backbone. The shared formulation is repeated next-frame prediction.

[minWM](#) adapts Wan2.1 directly; [Matrix-Game 2.0](#) uses the Wan architecture lineage.

Before you go: deadlines and office hour

OFFICE HOUR TODAY

4–5 PM

MC 4021 · this classroom

ASSIGNMENT 3

Tue Aug 4 · 11:59 PM

Waterloo time

CS 686 FINAL PROJECT

Wed Aug 5 · 11:59 PM

Report + code submission

Final exam—and good luck

SATURDAY AUG 8

7:30–10:00 PM

PAC 5

BRING

Non-programmable calculator

One double-sided letter-size notes sheet

This is our last class. Good luck!